

Shivam Kumar Thakur

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LinkedIn — GitHub — Portfolio

Professional Summary

Aspiring Cybersecurity and Technology Risk Professional currently pursuing M.Tech in Computer Science and Cybersecurity at IIT Patna. Strong foundation in Cybersecurity, Information Security, Network Security, Cloud Computing, AWS, Linux, and Software Development with hands-on experience in cloud automation, network optimization, routing enhancement, encryption systems, and AWS security services. Demonstrated leadership through technical clubs, public speaking, student mentoring, and technology events. Passionate about cyber resilience, technology risk management, secure digital infrastructure, and policy-driven innovation.

Education

Indian Institute of Technology (IIT) Patna 2026 – 2028

M.Tech in Computer Science and Engineering

NIST University, Berhampur, Odisha 2021 – 2025

Bachelor of Technology (Information Technology)

CGPA: 8.30 / 10

Gyansthali School, Gaya, Bihar (CBSE) 2021

Class XII – 71%

St. Thomas English School, Jharsuguda, Odisha (ICSE) 2019

Class X – 92%

Technical Skills

Programming Languages: Python, Java, C, SQL, HTML, CSS

Cybersecurity & Networking: Network Security, Information Security, Routing & Switching, Risk Management, Wireshark, Cisco Packet Tracer

Cloud Technologies: AWS (EC2, S3, WAF), GCP, IBM Cloud

Web Technologies: JavaScript, React.js, Next.js, Tailwind CSS

Tools: Git, GitHub, Linux, VS Code, IntelliJ IDEA, VirtualBox, Firebase, Figma

Core Areas: Cybersecurity, Cloud Computing, Technology Risk Management, Network Optimization, Data Encryption, Secure Systems Design

Research and Technical Projects

Information-Centric Network Optimization Project 2024

- Designed a network optimization framework to minimize routing hop count and improve packet delivery efficiency.

- Developed Python-based routing optimization mechanisms for network performance enhancement.
- Improved communication efficiency and reduced dependency on centralized servers.
- Completed under the guidance of Dr. Kumari Nidhi Lal, VNIT Nagpur.
- Copyright registered with Copyright Office, Government of India.
- AI implemented for predicting the history of searches with the same project under the guidance of Dr. Sudhir Ranjan Pattnaik, NIST university

Gesture-Based AWS Cloud Automation

2024

- Developed a gesture-controlled automation system using OpenCV, NumPy, Python and AWS.
- Integrated computer vision with AWS EC2 services using Boto3.
- Automated cloud instance start and termination operations.
- Reduced manual cloud administration effort significantly.

Image Encryption Service

2024

- Developed a secure image encryption platform using Triple DES cryptographic techniques.
- Implemented secure encryption and decryption workflows.
- Improved protection against brute-force and interception attacks.
- Demonstrated practical application of cryptography in information security.

Dynamic Mess Management System

2023

- Developed QR-code based attendance and token generation platform.
- Automated student attendance recording and mess entry verification.
- Reduced manual operational workload significantly.
- Maintained digital attendance records using database integration.

Teacher-Student Mobile Application

2023

- Developed Android application using Java and Kotlin.
- Enabled faculty information sharing and study material distribution.
- Improved communication and academic resource accessibility.
- Supported dynamic content updates for both teachers and students.

Internship Experience

VNIT Nagpur

June 2024 – July 2024

Intern

- Worked on networking and routing optimization research.
- Contributed to modernization of routing infrastructure.
- Developed techniques aimed at reducing server dependency.
- Assisted in implementation and evaluation of network efficiency improvements.

ByteIQ Analytics Pvt. Ltd.

2025

Intern

- Worked on AWS Web Application Firewall (WAF) implementation.

- Learned cloud security controls and access protection mechanisms.
- Assisted in monitoring and securing web application traffic.
- Gained practical exposure to AWS security services.

Intellectual Property

Copyright Registration (Reg. No.: SW-20011/2025)

Information-Centric Network Optimization and Routing Enhancement Project

Copyright Office, Government of India

Certifications

- Foundations of Cybersecurity – Coursera
- Networks and Network Security – Coursera
- Manage Security Risks – Coursera
- Demystifying Networking – NPTEL
- Programming in Java – NPTEL
- AWS Cloud with AI Workshop – LinuxWorld

Leadership and Positions of Responsibility

Cloud Computing Club, NIST University

Secretary and Treasurer

- Managed administrative and financial operations of the club.
- Coordinated technical workshops and cloud computing events.
- Collaborated with more than 40 students on technology initiatives.

Technical Event Organizer and Speaker

- Organized more than 10 technology-oriented events.
- Delivered sessions on Cloud Computing, Artificial Intelligence and Cybersecurity.
- Mentored students through technical awareness programs.

Achievements

- Runner-Up, Inter-College Coding Competition.
- Third Position, Intra-College Badminton Competition.
- Finalist, Rangmanch 2.0 Poetry Competition.
- Active contributor to technical communities and student development initiatives.

Languages

Hindi, English, Bhojpuri, Odia